



Ministry of Higher Education  
and Scientific Research

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University of Carthage

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**National Institute  
of Applied Sciences and  
Technology**



## Personal and Professional Project

Speciality: Industrial IT and Automation

Level: 4th Year

Topic:

Wearable Device for Hypo-Hyperthyroidism Early  
Diagnosis

*Prepared by :*

***Kouba CHORFI***

***Manel BEN HASSINE***

***Aya ADDALI***

*Supervisor:*

***Mohamed Ali HAMDI***

School Year: 2024/2025

## Abstract

This paper presents the development of a low-cost, wearable smart sensing system designed for the early detection of thyroid-related abnormalities through non-invasive monitoring of key physiological parameters. The system integrates photoplethysmography (PPG) and temperature sensors to measure heart rate, blood oxygen saturation ( $\text{SpO}_2$ ), and skin surface temperature, which are analyzed to identify symptoms associated with hypothyroidism and hyperthyroidism. A machine learning model, trained on a synthetic health metrics dataset, classifies thyroid conditions based on predefined clinical thresholds. The hardware prototype, built around an ESP32 microcontroller, wirelessly transmits data to a web-based dashboard for real-time visualization and alerts. Despite limitations such as the lack of clinically annotated datasets and susceptibility to motion artifacts, the system demonstrates promising results in classifying thyroid dysfunction under controlled conditions. Future work aims to enhance diagnostic accuracy by incorporating ECG features, improving real-time data streaming, and integrating contextual questionnaires. This research contributes to the growing field of wearable health technologies by offering a scalable, cost-effective solution for early thyroid disorder detection, potentially reducing healthcare costs and improving patient outcomes.

**Keywords:** Wearable device, thyroid disorder detection, photoplethysmography (PPG), machine learning, real-time monitoring, non-invasive sensing.

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# 1 Project Context:

## 1.1 Background:

By definition provided on the National Cancer Institute on [1] is a gland located beneath the larynx (voice box) that makes thyroid hormone and calcitonin. The thyroid gland helps regulate growth and metabolism.

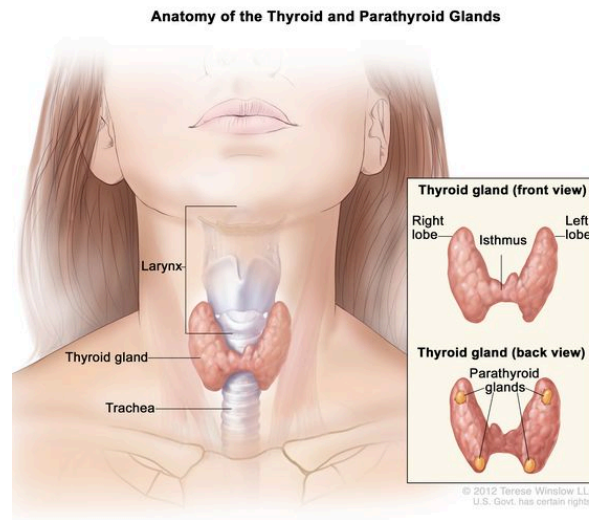


Figure 1: Anatomy of the Thyroid gland

Thyroid dysfunction occurs when the gland produces either excess hormones (**hyperthyroidism**) or insufficient levels (**hypothyroidism**). Also, it presents itself as cancer, swelling or lumps in the gland. [2]

Irregularities in thyroid hormone production can lead to significant physiological abnormalities, particularly in cardiovascular, neurological, and reproductive systems. The incidence of thyroid disorders has been increasing in recent years, with a substantial number of individuals diagnosed annually. [3]

At least a person out of ten is suffering from thyroid disease in India. The disorder of thyroid disease primarily happens in women aged 17–54. The extreme stage of thyroid results in cardiovascular complications, increase in blood pressure, maximizes the cholesterol level, depression and decreased fertility. [4]

Thyroid disorders are a significant public health concern, affecting a substantial portion of the population. In the U.S., more than 12% of individuals will develop a thyroid condition during their lifetime, with an estimated 20 million Americans currently living with some form of thyroid

disease. Alarming, up to 60% of those affected remain undiagnosed, increasing their risk of serious complications such as cardiovascular disease, osteoporosis, and infertility.

Women are disproportionately impacted, being five to eight times more likely than men to develop thyroid problems, and one in eight women will experience a thyroid disorder in her lifetime. While most thyroid cancers are treatable, a small percentage exhibit aggressive behavior.

The exact causes of thyroid dysfunction remain largely unknown, underscoring the need for greater awareness and early detection. Undiagnosed or poorly managed hypothyroidism in pregnant women poses additional dangers, including higher risks of miscarriage, preterm birth, and impaired child development. Despite these challenges, the majority of thyroid conditions are chronic yet manageable with appropriate medical care, highlighting the importance of timely diagnosis and treatment. [5]

Common symptoms of thyroid disorders vary depending on the type. In hypothyroidism, symptoms may include fatigue, weight gain, cold intolerance, depression, dry skin, constipation, and slowed heart rate. In contrast, hyperthyroidism is often associated with weight loss, heat intolerance, anxiety, increased heart rate, sweating, tremors, and sleep disturbances. These symptoms can easily be confused with other health issues, which contributes to the high number of undiagnosed cases.

## 1.2 Objective:

In order to provide a better life for individuals and avoid undiagnosed cases, over-checking, and paranoia about getting diseases, especially among women, the cost of labs and ultrasound scans is significantly increasing in the healthcare industry. Monthly check-ups will cost at least 200 TND in Tunisia for one person, including the Labs, and 50 TND for an ultrasound.

Our goal is to lower these expenses, reduce the psychological stress associated with health concerns, and support early detection of thyroid disorders — particularly for people who may otherwise go undiagnosed.

The primary objective of this project is to develop a low-cost smart sensing system capable of detecting changes in skin surface temperature and cardiovascular rhythms through a non-invasive method for the early diagnosis of thyroid-related abnormalities.

### 1.3 Adopted Solution:

We propose a wearable device prototype designed to detect symptoms of hypothyroidism or hyperthyroidism in users and provide alerts along with a brief questionnaire. This self-assessment helps users determine whether a medical consultation is necessary. If no risk is detected, users can avoid unnecessary anxiety and save on healthcare expenses.

Common symptoms include feeling excessively cold (typically associated with hypothyroidism) or excessively hot (associated with hyperthyroidism), due to the thyroid gland's impact on body temperature regulation. When combined with other vital signs such as blood oxygen saturation (SpO<sub>2</sub>) and heart rate, we can obtain a preliminary understanding of the user's thyroid condition.

To gather this data, we use a photoplethysmograph (PPG) sensor, which is a non-invasive tool commonly used to measure blood flow volume and evaluate the physiological condition of a person. By analyzing the PPG signals, our system can monitor both heart rate and SpO<sub>2</sub> levels continuously and accurately. In future iterations, we aim to enhance this further by extracting ECG-like features from PPG data through advanced signal processing.

Our goal is to bring some of that precision into a simpler, wearable form using enhanced signal processing on the PPG data. This would allow the device to better detect irregularities in heart rate that may be linked to thyroid disorders, all without the need for complex or expensive hospital tools.

A dashboard will serve as the Graphical User Interface (GUI), displaying real-time readings of temperature, heart rate, and SpO<sub>2</sub>. If any abnormal readings are detected, the system will trigger an alert message for the user, guiding them toward further consultation if needed.

## 2 Principle:

### 2.1 PPG:

Photoplethysmography (PPG) is a non-invasive optical technique used to investigate subdermal perfusion. Measurements performed at the skin surface detect changes in blood volume in the microvascular bed (Figure 2). PPG is used to monitor vital parameters such as pulse rate, arterial blood oxygen saturation (SpO<sub>2</sub>) and/or respiratory rate. [6]

A PPG typically consists of a light source and a photodetector. Devices using PPG technology often rely on LEDs of various wavelengths, such as green, red, and blue light. The amount of energy that reaches the skin depends on the wavelength of the light; for instance, green light is commonly used because it offers a high signal-to-noise ratio (SNR). These colored light signals enable the device to determine both pulse rate and respiratory rate. [7]

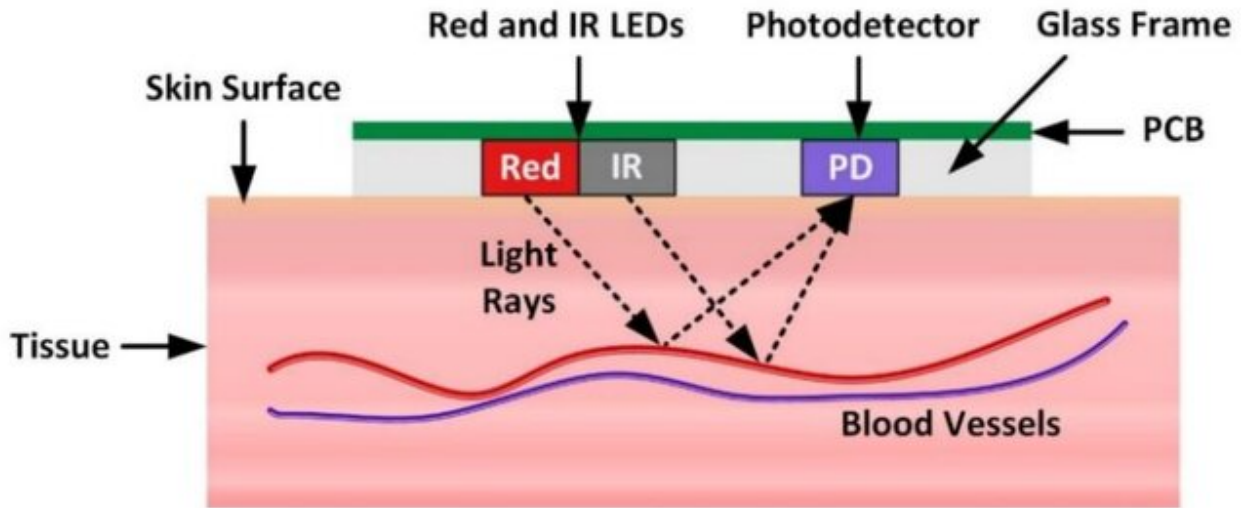


Figure 2: Working principle of a reflection-type PPG sensor. [8]

Figure 3 shows the detection of a photoplethysmographic transmission waveform, which has direct current (DC) and alternating current (AC) components. The DC component of the PPG waveform corresponds to the transmitted or reflected optical signal from the tissue and depends on the structure of the tissue and the average volume of both the arterial and venous blood. The DC component changes slowly with respiration, while the AC component fluctuates according to the changes in blood volume that occur between the systolic and diastolic phases of the cardiac cycle. The fundamental frequency of the AC component depends on the heart rate (HR) and is superimposed onto the DC component. [7]



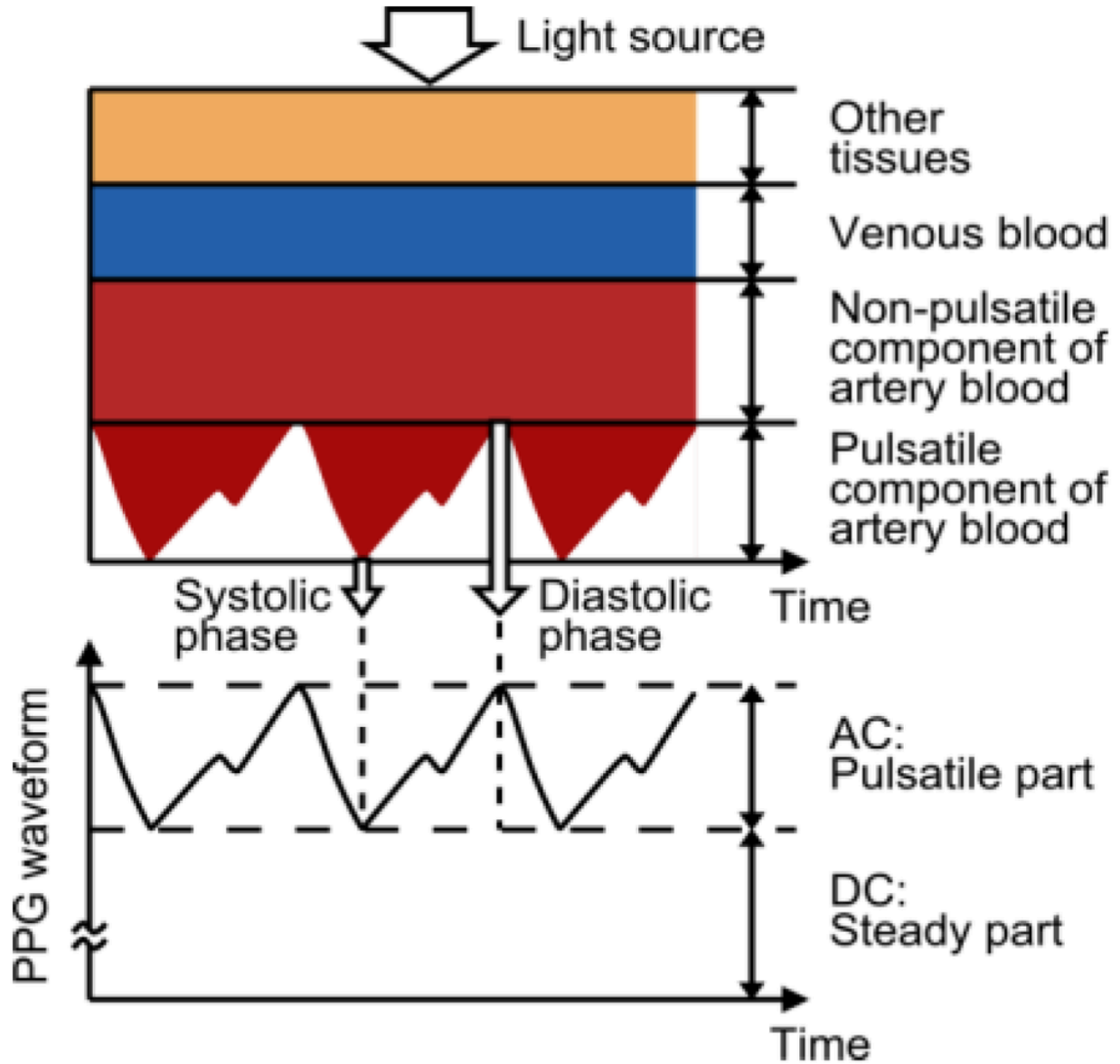


Figure 3: Variation in light attenuation by tissue. [9]

Our PPG (photoplethysmography) sensor utilizes the MAX30102, an integrated optical module containing a red LED (660 nm) and an infrared LED (880 nm), along with a high-sensitivity photodiode. The LEDs emit light that is absorbed and reflected by blood vessels and tissue, while the photodiode captures the reflected signal. The MAX30102 features an 18-bit ADC, programmable sample rates (50–3200 Hz), ambient light cancellation, and adaptive LED current control (up to 50 mA). Its compact size (5.6 mm × 3.3 mm), low power consumption (<1 mW in HR mode), and built-in FIFO for data storage make it ideal for wearable health monitoring. We leverage its motion artifact resilience and high SNR to extract accurate heart rate and SpO<sub>2</sub> measurements.[10]

## 2.2 Heart Rate:

In medicine, the pulse represents the tactile arterial palpation of the heartbeat by trained fingertips. Resting heart (pulse) rate (RPR) is a potentially useful measure for neurological fitness calculated as beats per minute (bpm). The normal RPR ranges for two different age groups are a) ages 6-15 (70-100 bpm) and b) 18 and over (60-100 bpm). [3] [11]

When your resting heart rate (HR) is over 100 bpm, providers call this tachycardia. This means you may have an issue with your heart's conduction system that signals your heart to beat. Examples of this type of issue include atrial flutter and ventricular tachycardia. And this may indicate a Thyroid disorder. [12]

Thyroid hormones help control heart rate. In people with clinical hypothyroidism, the heart rate is often 10–20 (bpm) slower than usual. [13] [14]

The opposite problem, hyperthyroidism, or too much thyroid hormone, is far less common, affecting less than 1% of the population. But it, too, can harm the heart. [15] Sinus tachycardia, defined as a HR of >100 bpm, is another common sign of hyperthyroidism in 70–100% of patients. [16] [17] [18]

In a simple environment to get our HR, when you feel your pulse, count the number of beats in 15 seconds. Multiply this number by four to calculate your beats per minute. [19]

Regarding PPG heart rate detection, there are two main approaches used: time-domain sophisticated filtering and frequency-domain processing. Both are often accompanied by a movement sensor (accelerometer) for movement-based artefacts/spectrum removal.

A straightforward approach toward heart rate estimation is peak detection in a periodic signal. One of the simplest possibilities is to use threshold or auto-threshold values in the signal time window. Another way is to use transforms such as continuous wavelet or Hilbert. Also, others, such as, a nonlinear filter bank was used, with variable cutoff frequencies, and the detection algorithm was based on a time-varying autoregressive model, with a Kalman filter used for autoregressive parameter estimation. [20]

In our work, the detection of heart rate will be detected using the time-domain heart rate measurement algorithm (TDHR) consists of three main blocks: signal conditioning, peak detection, and heart-rate-measuring blocks.

As mentioned [20], all of the signal processing was done while taking into account the need for low power, low resources, and computing power utilization, necessary for a self-sufficient mobile sensor.

## 2.3 SpO<sub>2</sub>:

In a PPG, the oxygen saturation of arterial blood (SpO<sub>2</sub>%) is calculated from the amplitude of transmitted R and IR light, and the PR is determined from the PPG wave. [7]

Hemoglobin (Hb) is the protein in red blood cells (RBCs) responsible for transporting oxygen. It exists primarily in two forms: oxygenated hemoglobin (oxyhemoglobin, HbO<sub>2</sub>) and deoxygenated hemoglobin (deoxyhemoglobin, RHb).

Peripheral capillary oxygen saturation (SpO<sub>2</sub>) refers to the estimated percentage of hemoglobin in the capillaries that is saturated with oxygen, calculated as the ratio of oxyhemoglobin to total hemoglobin.

$$SpO_2 = 100 \times \frac{C[HbO_2]}{C[HbO_2] + C[RHb]} \quad (2.3.1)$$

where C[HbO<sub>2</sub>] and C[RHb] are the concentrations of HbO<sub>2</sub> and RHb, respectively. [21]

A detailed explanation of the science behind the PPG LED detection and HbO<sub>2</sub> and RHb appearance in the blood in [21] and [7], but we are going to proceed with is the idea of PPG is based on two principles: modulation of transmitted light by absorption of pulsatile arterial blood and different absorption characteristics of HbO<sub>2</sub> and RHb for different wavelengths.

For SpO<sub>2</sub> measurement with our Sensor Max30102, we are using the Calibrating Maxim's SpO<sub>2</sub> Algorithm:

$$SpO_2 = aR^2 + bR + c \quad (2.3.2)$$

Where R is determined by

$$R = \frac{AC_{red}/DC_{red}}{AC_{ired}/DC_{ired}} \quad (2.3.3)$$

a, b, and c are calibration coefficients.

We use the default coefficients of the MAX30101—the closest prototype to our sensor—to obtain medically accurate values: a = 1.5958422, b = -34.6596622, and c = 112.6898759. This

calibrated equation enhances estimation accuracy compared to generic models and aligns with manufacturer-recommended practices for prototype-level pulse oximetry.

One of the roles of thyroid hormones is to promote oxygen delivery to body tissues. Therefore, low thyroid hormone levels, as observed in hypothyroidism, may result in reduced tissue oxygenation [22]. Normal pulse oximeter readings for peripheral oxygen saturation ( $\text{SpO}_2$ ) in children and adults typically range from 95% to 100%. Values below 93% may indicate inadequate oxygen levels. [23]

A human study examining acute hyperthyroidism found that increasing circulating thyroid hormone levels did not significantly affect skeletal muscle oxygenation or blood flow [24]. Therefore, the impact of hyperthyroidism on oxygen saturation remains uncertain and will not be considered further in our detection analysis.

## 2.4 Temperature:

Body temperature is not a primary indicator of thyroid disorders. While hypothyroidism can slow metabolism and lead to a decrease in core body temperature—causing individuals to feel cold or have low cold tolerance [25]—hyperthyroidism is often associated with increased sweating and sensitivity to heat [26]

However, a large-scale study involving 35,488 patients, cited by the British Thyroid Foundation [27], reported an average body temperature of  $36.6^\circ\text{C}$ . Notably, 95% of the population had a body temperature between  $35.7^\circ\text{C}$  and  $37.3^\circ\text{C}$ . This implies that if  $37^\circ\text{C}$  were used as a strict threshold, more than three-quarters of the normal population would be misclassified as hypothyroid.[28]

Therefore, while temperature will be monitored by our wearable device, it will not be used as a definitive diagnostic indicator. For reference, we will set  $36.4^\circ\text{C}$  as the lower threshold suggestive of hypothyroidism and  $37.3^\circ\text{C}$  as the upper threshold suggestive of hyperthyroidism.

## 3 System Design

### 3.1 Hardware Overview:

The main objective of this system is to measure body temperature, heart rate, and  $\text{SpO}_2$  using an ESP32-based IoT device, detect possible thyroid-related abnormalities, and display the readings on a web dashboard with alerts if symptoms are detected (Figure 4).

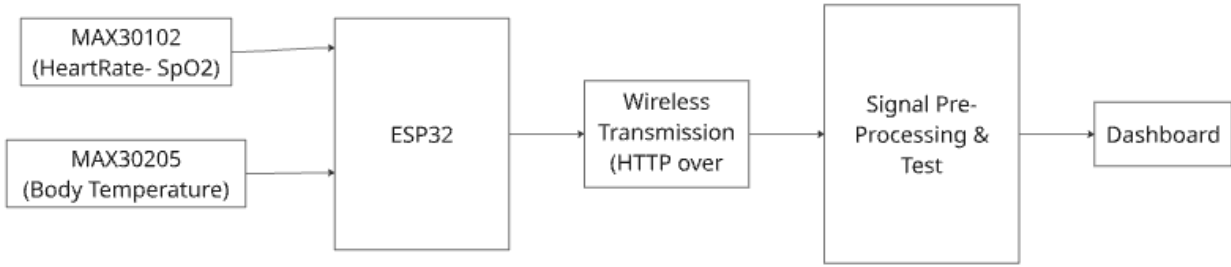


Figure 4: Block Diagram of the proposed non-invasive thyroid detection

### 3.1.1 Sensors & MicroController:

To monitor key physiological indicators relevant to thyroid dysfunction, two biomedical sensors were integrated into the system. The MAX30102 sensor is employed for non-invasive measurement of heart rate and blood oxygen saturation (SpO<sub>2</sub>) using photoplethysmography (PPG). This sensor integrates red and infrared LEDs along with a photodetector, offering high sensitivity in detecting blood volume changes. Its compact form factor and low power consumption make it particularly suitable for wearable health monitoring applications, where both accuracy and efficiency are required.

In parallel, the MAX30205 sensor is used to measure skin temperature with medical-grade accuracy ( $\pm 0.1^{\circ}\text{C}$ ). It operates over the I<sup>2</sup>C protocol and is certified for clinical use, ensuring reliable readings. Since abnormal thyroid hormone levels can influence thermoregulation, manifesting as low tolerance to cold in hypothyroidism or increased sensitivity to heat in hyperthyroidism, body temperature serves as an important auxiliary parameter in thyroid assessment. Thus, combining both sensors allows for a more comprehensive evaluation of potential thyroid-related symptoms.

The ESP32 microcontroller was selected as the central processing unit due to its integration of Wi-Fi and Bluetooth capabilities, which facilitate real-time communication with a remote server or local dashboard without the need for additional communication modules. Its dual-core processor and low-power architecture make it ideal for continuous operation in wearable, battery-powered systems. Moreover, the ESP32 supports multiple GPIOs and I<sup>2</sup>C communication, enabling concurrent interfacing with both the MAX30102 and MAX30205 sensors. This allows seamless data acquisition and sensor control. Additionally, the ESP32 is capable of lightweight on-device signal preprocessing and supports HTTP communication, which simplifies data transmission to cloud-based platforms or web-based dashboards for further analysis and visualization.

### 3.1.2 Circuit Design:

The circuit integrates two biomedical sensors—the MAX30102 and the MAX30205 (Figure 4.1)—with the ESP32-WROOM microcontroller. The MAX30102 is used for measuring heart rate and blood oxygen levels, while the MAX30205 provides high-precision body temperature readings. These sensors were selected for their accuracy and compatibility with low-power embedded systems. The ESP32-WROOM, featuring built-in Wi-Fi and low energy consumption, serves as the central processing unit, enabling real-time data processing and wireless communication.

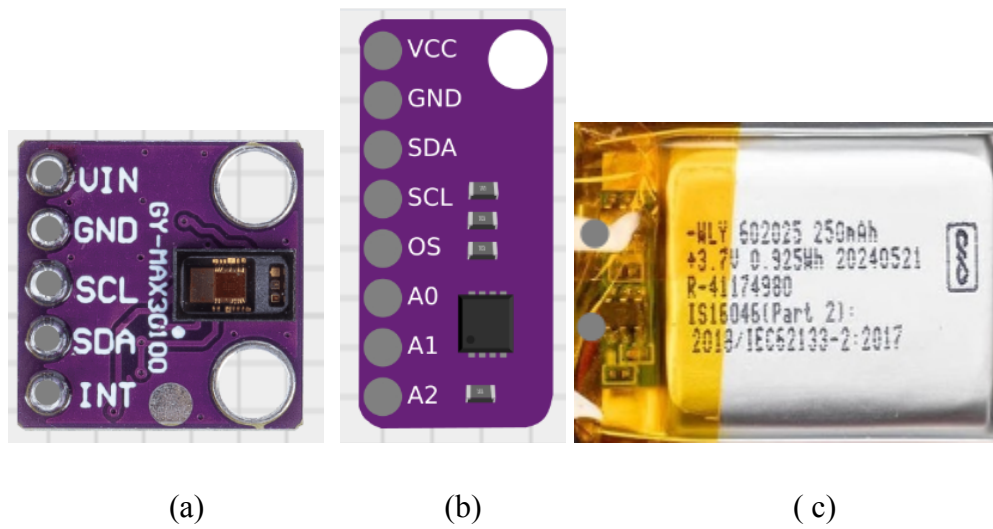


Figure 4.1: (a) Max30102 sensor | (b) Max302005 sensor | (c ) 3.7V LiPo Battery

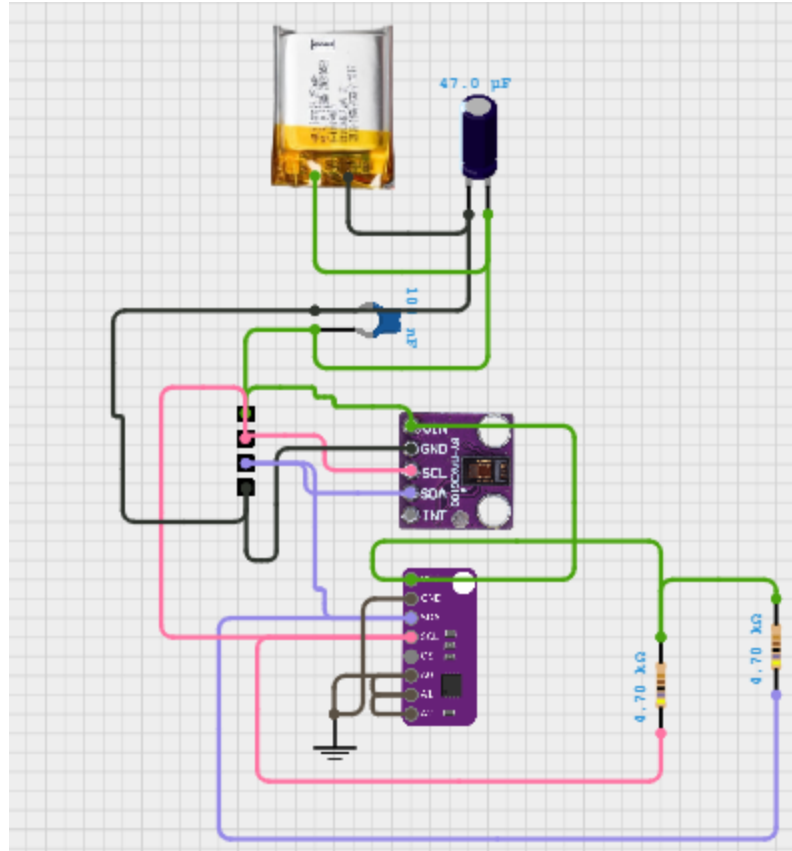


Figure 4.2: Circuit Wiring Prototype using Cirkit Design ( Black Wire: GND | Green Wire: Vin | Purple Wire: D22 | Pink: D21)

The schematic (Figure 4.2), created using Cirkit Design, illustrates the interconnections between components via the I<sup>2</sup>C protocol. The MAX30205 is configured to use the address 0x48 by grounding pins A0, A1, and A2, while the MAX30102 operates at address 0x56.

Due to time constraints, a custom PCB was not developed at this stage; instead, a functional prototype was assembled on a pegboard for easier testing and modification. For power, a 3.7V 250mAh Lithium Polymer (LiPo) battery was selected, providing safe and sufficient voltage within the operational limits of both sensors (ranging from -0.3V to +6.0V for the MAX30102 and up to +4.0V for the MAX30205).

Additionally, a Type-C USB charging module was prototyped (Figure 4.3) and will be integrated later, pending further testing to ensure overcharging protection. This setup offers a reliable, compact, and low-power platform for physiological signal monitoring, with plans to transition to a custom PCB for improved integration and robustness.

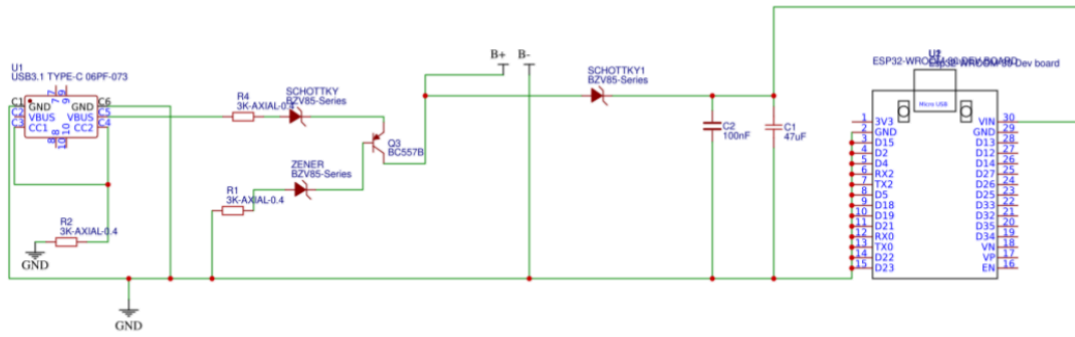


Figure 4.3: Charging Module Prototype design

### 3.1.3 Communication & Display Interface:

To facilitate data transmission and visualization, the system employs the HTTP protocol over Wi-Fi, enabling seamless communication between the ESP32 microcontroller and a remote server or local dashboard. This communication method was selected due to its simplicity, efficiency, and broad compatibility with modern web technologies.

A web-based graphical user interface (GUI) was developed to display real-time readings of heart rate, SpO<sub>2</sub>, and body temperature. The dashboard incorporates embedded logic for preliminary condition detection, such as flagging potential hypothyroidism when SpO<sub>2</sub> or temperature readings fall below defined clinical thresholds. In cases where sensor data suggests abnormal patterns associated with thyroid dysfunction, alert notifications are triggered to prompt user attention.

The graphical user interface (GUI) is developed using standard web technologies—HTML, CSS, and JavaScript—for the Frontend alongside Flask with Python for the Backend, and can be hosted locally as a prototype.

## 3.2 Detection Workflow:

The main objective of this application is to measure body temperature and pulse rate continuously throughout the day to detect the type of thyroid disorder. The system uses the MAX30102 sensor to measure pulse rate and SpO<sub>2</sub>, and the MAX30205 sensor for accurate body temperature measurement.

Data for three key parameters—temperature, heartbeat, and SpO<sub>2</sub>—are collected throughout the day, and the daily mean values are calculated before being analyzed by the classification model. If HR and temperature mean values exceed their respective normal thresholds, the system indicates hyperthyroidism; if all three are below normal thresholds, it indicates hypothyroidism.



If the data does not consistently satisfy either condition, the system classifies the thyroid status as normal. (Figure 5)

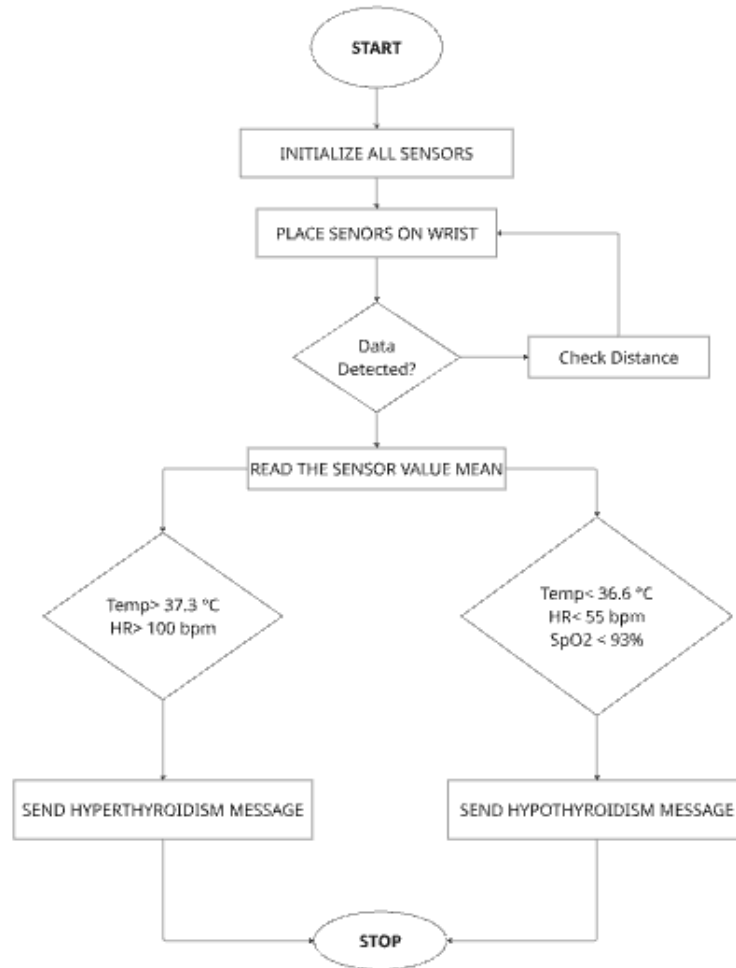


Figure 5: Flow chart of Thyroid level detection

We are acquiring vital physiological signals using sensors, specifically the MAX30102. For optimal signal quality and power efficiency, we set the Red LED amplitude to 0x50 and the IR LED amplitude to 0x4F. These values correspond to the LED pulse amplitude current in microamps ( $\mu\text{A}$ ). According to the MAX30102 datasheet [10], the current ranges from 0x00 (0  $\mu\text{A}$ ) to 0xFF (50.0 mA), with each increment representing approximately 196  $\mu\text{A}$ . Setting the Red and IR LEDs just one step apart ensures that both operate at nearly the same current, approximately 15.5–15.7 mA, balancing signal quality and power consumption.

### 3.3 Software Implementation:

#### 3.3.1 Signal Processing:

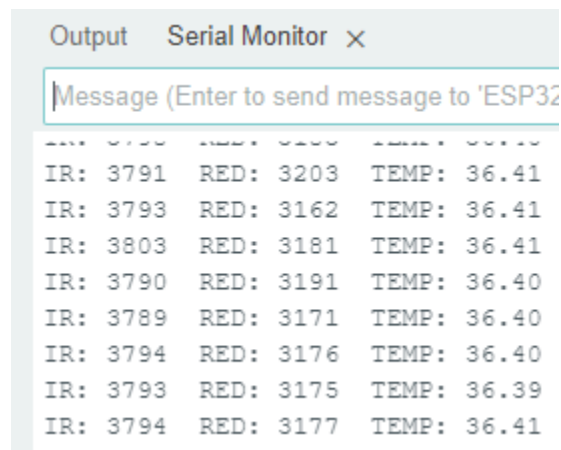
The ESP32 microcontroller communicates with the MAX30102 pulse oximeter and the MAX30205 temperature sensor over I<sup>2</sup>C using the *Wire.h* library.

The MAX30102 is configured via *MAX30105.h* to operate with red and Infrared (IR) LED pulse amplitudes set to 0x50 and 0x4F, respectively, enabling photoplethysmographic acquisition at a sampling rate of approximately 100 Hz (limited in this case to ~1 Hz by loop delay for demonstration and bandwidth control).

The MAX30205, offering a resolution of 0.0039°C and accuracy of ±0.1°C (clinical range), is initialized with default settings.

Networking is managed using *WiFi.h* and *WebServer.h*, allowing the ESP32 to serve a local HTTP endpoint (/data) at port 80. Sensors' data (IR wavelength detected, RED wavelength detected, temperature) was encapsulated in a JSON payload and made accessible in real-time. This configuration provides a low-overhead, real-time physiological monitoring pipeline, suitable for integration in embedded or IoT-based health systems.

The result will be sent as a JSON payload shown in Figure 6.

The image shows a screenshot of the Arduino IDE Serial Monitor window. The title bar says "Output Serial Monitor" with a close button. Below the title bar is a text input field with the placeholder "Message (Enter to send message to 'ESP32...')". The main area of the window displays a stream of sensor data in a monospaced font. The data is organized into three columns: IR, RED, and TEMP. Each row represents a single data point. The IR values range from 3789 to 3803. The RED values range from 3162 to 3203. The TEMP values range from 36.39 to 36.41.

| IR   | RED  | TEMP  |
|------|------|-------|
| 3791 | 3203 | 36.41 |
| 3793 | 3162 | 36.41 |
| 3803 | 3181 | 36.41 |
| 3790 | 3191 | 36.40 |
| 3789 | 3171 | 36.40 |
| 3794 | 3176 | 36.40 |
| 3793 | 3175 | 36.39 |
| 3794 | 3177 | 36.41 |

Figure 6: Serial Monitor Results of Arduino IDE Upload

The physiological data (IR, RED, and temperature) is acquired from an ESP32 microcontroller via HTTP GET requests to a local endpoint, with a sampling rate of 100 Hz, which is appropriate for capturing detailed photoplethysmographic (PPG) signals. The signal buffers are configured with a capacity of 200 samples, providing approximately 2 seconds of data per cycle.

Visualization and processing are updated every 100 ms (i.e., 10 Hz), balancing real-time responsiveness with computational efficiency.

Each update involves band-pass filtering (0.5–4.0 Hz) to isolate cardiac-related oscillations. According to [29], the pulsatile component of the PPG lies within 0.5–4 Hz, while motion artifacts (MAs) generally occur below 0.5 Hz or above 6 Hz. Therefore, a band-pass filter of 0.5–6 Hz was selected to suppress baseline drift and high-frequency noise while preserving the essential signal components.

For HR, following the three stages of our TDHR algorithm. After filtering, the first derivative of the IR signal is taken to highlight rising edges corresponding to cardiac pulses. Local maxima in the derivative that exceed a threshold—proportional to the derivative's standard deviation—are identified as candidate peaks. A debouncing mechanism ensures a minimum inter-beat interval equivalent to a maximum of 240 BPM to eliminate false positives. BPM is then calculated by dividing the number of valid peaks by the signal duration:

$$\text{BPM} = \left( \frac{N_{\text{peaks}}}{T_{\text{seconds}}} \right) \times 60 \quad (3.3.1)$$

with N: number of detected heartbeats in peaks, T: time window in seconds

For SpO<sub>2</sub>, using equation (2.3.3), we computed the ratio of AC (pulsatile) to DC (baseline) components for both infrared (IR) and red PPG signals. The AC component is extracted using the root mean square (RMS), while the DC component is calculated as the mean of the raw signal. This value is then mapped to oxygen saturation using an empirically derived second-order polynomial in equation (2.3.2) with the mentioned calibration coefficient values.

The result is constrained to a physiological range of 70% to 100% to mitigate error from transient anomalies for blood oxygen saturation. This method offers real-time HR and SpO<sub>2</sub> estimation with minimal computational load, ideal for embedded systems. (Figure 7)

To ensure accurate signal acquisition, the sensor must be positioned at a distance between 5 mm and 15 mm from the skin surface. Deviations from this range may introduce spaced fluctuations and signal instability, which are slightly introduced in our plots.

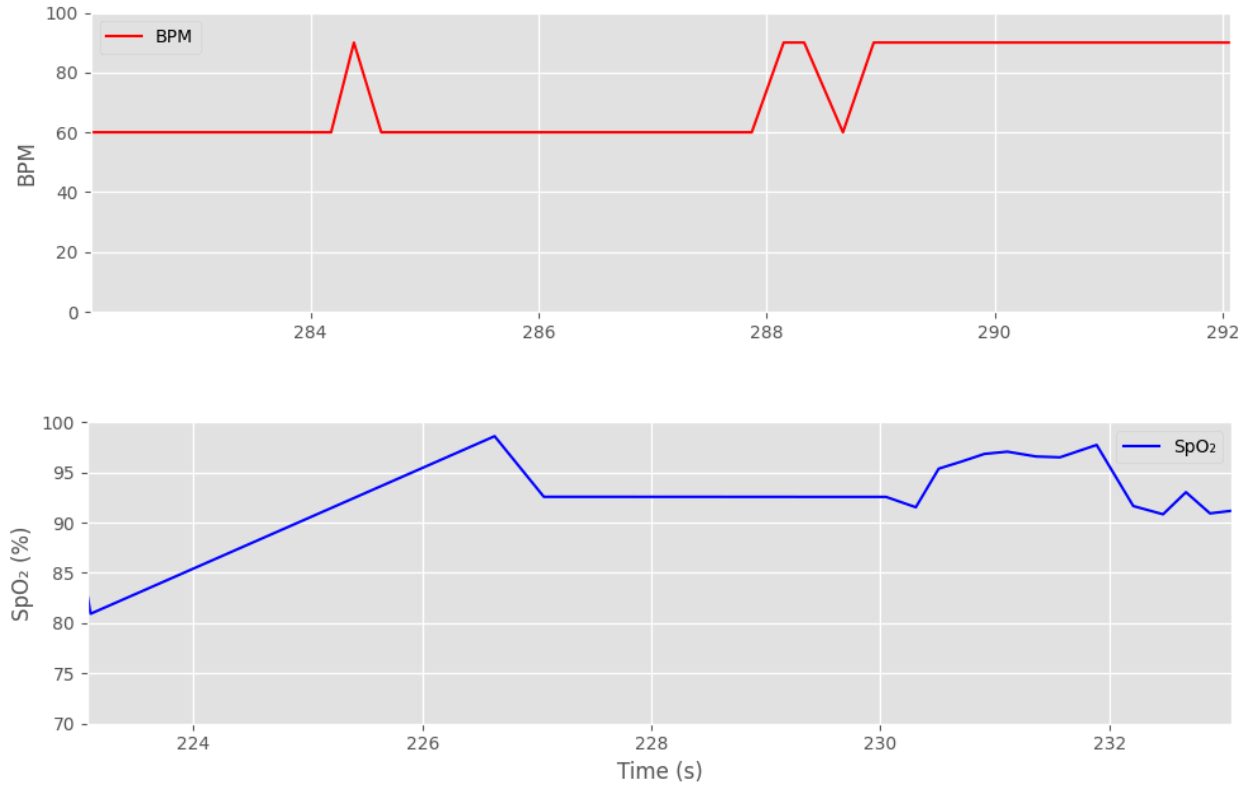


Figure 7: Time-series plots of heart rate (BPM) and SpO<sub>2</sub> (%)

### 3.3.2 GUI Dashboard:

As part of the real-time monitoring system, a user-friendly web dashboard (Figure 8) was developed to facilitate the visualization of physiological data acquired from a wearable sensor platform. The backend is implemented using the Flask microframework (Python), while the frontend leverages standard web technologies, including HTML, CSS, JavaScript, and the Bootstrap framework for responsive design.

The dashboard presents three key biometric parameters—heart rate, blood oxygen saturation (SpO<sub>2</sub>), and body temperature—captured by the wearable system. Although the sensor transmits data via Wi-Fi, the current architecture operates in a semi-real-time configuration. Specifically, processed physiological data is saved to a CSV file following signal analysis in Python.

Upon client access, the dashboard parses this file and renders both instantaneous measurements and historical trends through time-series plots. This dual-layered view supports both momentary assessment and retrospective evaluation of patient status. The chosen architecture is tailored for the development and testing phases, during which data is processed at discrete intervals (e.g.,

every few minutes or hours), rather than streamed continuously. This design simplifies debugging, ensures reproducibility, and facilitates post hoc analysis of alert conditions.

The server exposes two primary API endpoints:

- `/csv-data`: returns the full dataset, serialized in JSON format;
- `/live-data`: accepts a timestamp parameter and returns only entries recorded since the last query, supporting incremental data retrieval.

Efficient data access is ensured via helper functions that selectively read and parse the CSV file to retrieve either the latest entry or the complete series. This avoids redundant I/O operations and reduces computational overhead.

The frontend interface employs Plotly.js for high-resolution, interactive visualizations. Upon initialization, the dashboard loads historical data to populate the time-series charts for heart rate, SpO<sub>2</sub>, and temperature. Subsequently, it enters a polling loop using JavaScript's `setInterval()`, issuing asynchronous requests to `/live-data` at one-second intervals to maintain near real-time responsiveness.

The dashboard is hosted locally and accessed via the machine's IP address. To update the charts with the most recent data, the user simply refreshes the browser page, which triggers a new read of the underlying CSV file and updates all visualizations accordingly.

Figure 7 presents the dashboard interface, which displays real-time status indicators for vital signs, along with dynamic warnings or alerts when values exceed clinically defined thresholds. When the machine learning model detects patterns indicative of hypothyroidism or hyperthyroidism, a corresponding diagnostic message is generated and shown in the "Trends and Alerts" section.

It is important to note that temperature measurements may exhibit temporary deviations during initial sensor warm-up. This is a known behavior of the MAX30205 sensor, which requires a short stabilization period to yield accurate readings. During this phase, the system was able to correctly identify a hypothyroid pattern, validating the model's ability to detect low-temperature anomalies. However, for consistent and reliable results in real-world deployment, vital sign acquisition should be performed once the sensor reaches thermal equilibrium to avoid misinterpretation of transient values.

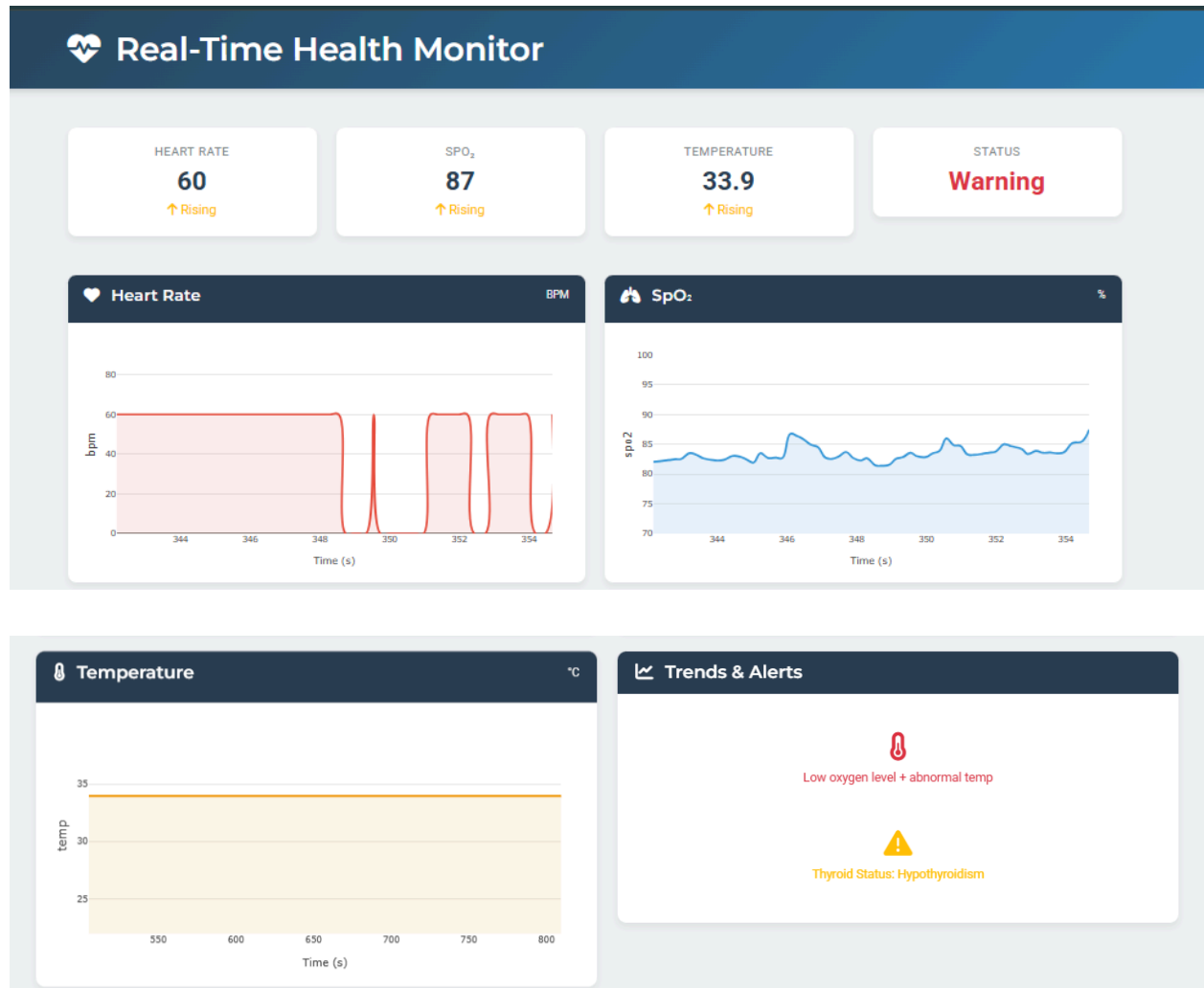


Figure 8: Web dashboard displaying heart rate, SpO<sub>2</sub>, and temperature data with real-time trends and alerts.

## 4 ML Model Development and Training:

To enable accurate detection of thyroid gland disorders, we adopted a machine learning-based classification model trained with diverse categorical limits. The classification aims to evaluate the patient's condition—such as treatment status, general health state, and presence of thyroid-related anomalies—to facilitate early diagnosis and timely intervention. While various machine learning (ML) and deep learning (DL) approaches have been applied to thyroid disease prediction, comprehensive evaluations and performance comparisons across techniques remain underexplored in the current state of the art. [30]

Our proposed methodology builds upon the framework presented in [30], with modifications due to data availability constraints. Specifically, we employed an alternative dataset, as vital signal recordings from clinically diagnosed euthyroid, hypothyroid, and hyperthyroid patients were not accessible. In the absence of publicly available datasets containing physiological signals explicitly linked to thyroid disorders, we selected alternative datasets that exhibit analogous physiological variability to simulate thyroid-related conditions.

The system architecture is illustrated in Figure 9. The pipeline begins with dataset labeling, from which specific target classes are selected to define the classification objective. To mitigate issues related to class imbalance, a data balancing technique is applied prior to training. The preprocessed dataset is then partitioned into training and testing subsets. Feature selection is conducted to enhance model generalization and minimize overfitting. Subsequently, the classification model is trained on the processed data and validated using the reserved test set. The model outputs categorical predictions corresponding to thyroid states (e.g., hypothyroid, hyperthyroid, or normal), and performance is evaluated using standard classification metrics.

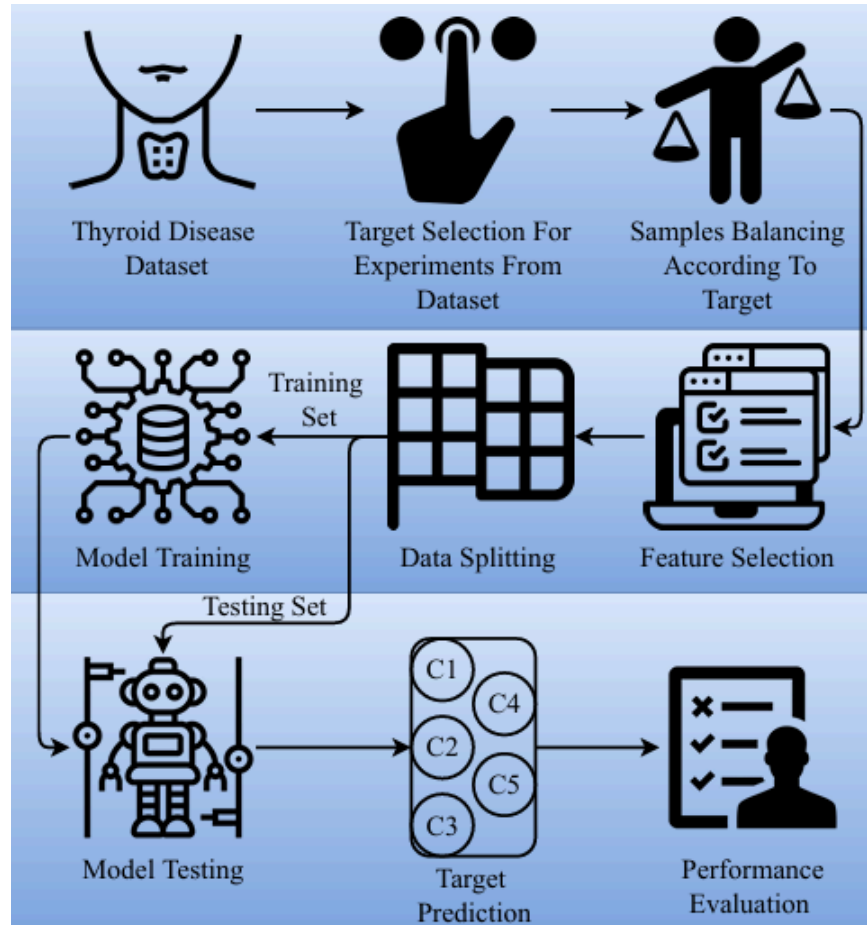


Figure 9: Flow of the proposed methodology

## 4.1 Dataset Preparation and Classification Strategy

### 4.1.1 Dataset Description

Due to the lack of publicly available datasets dedicated specifically to thyroid-related vital sign detection, we focused on identifying a dataset that captures fluctuations in a range of vital parameters. We selected the dataset presented in [31], which consists of momentary vital sign readings along with a binary label indicating whether the patient is Positive or Negative for a thyroid-related condition.

The dataset comprises synthetic health metrics for 10,000 individuals, including key physiological parameters such as heart rate, blood pressure, and SpO<sub>2</sub> levels (Figure 10). These features offer sufficient granularity to support rule-based relabeling, allowing for the derivation of meaningful patterns and the development of robust machine learning models for thyroid disorder classification.

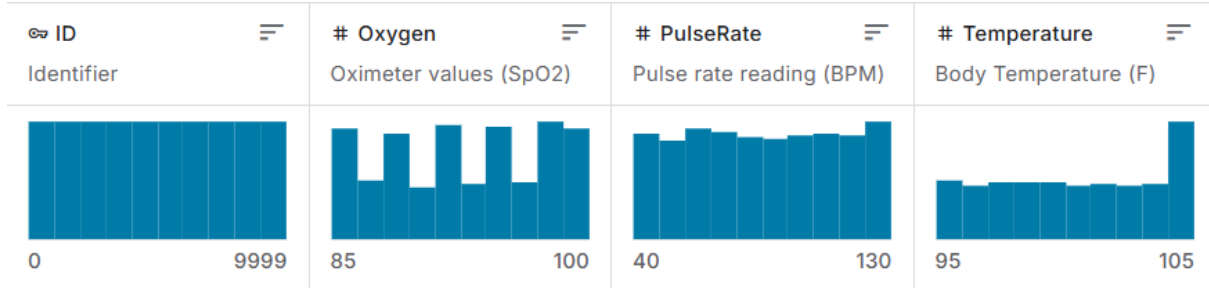


Figure 10: Details of the dataset

### 4.1.2 Data Loading and Initial Processing

After importing the dataset from the CSV file, we focused on three clinically relevant health indicators: body temperature, oxygen saturation (SpO<sub>2</sub>), and heart rate, as these variables are commonly associated with thyroid dysfunction. An initial feature engineering step involved converting body temperature values from Fahrenheit to Celsius using the standard conversion formula:

$$T(^{\circ}C) = (T(^{\circ}F) - 32) \times \frac{5}{9} \quad (4.1.2)$$

This ensured consistency with international medical reporting standards.

To validate the dataset and ensure the reliability of these features, a preliminary statistical analysis was conducted using the 'df.describe()' method. This step enabled verification that the



values of all selected features fell within expected physiological limits. The summary statistics also provided insights into potential outliers or anomalies that might require further handling during subsequent preprocessing stages.

Outlier detection and basic statistical profiling helped ensure data consistency before model development. The dataset showed realistic and physiologically plausible values for body temperature, SpO<sub>2</sub>, and heart rate, confirming its reliability for subsequent analysis. (Figure 11)

|       | Body Temperature | SpO2         | Heart Rate   |
|-------|------------------|--------------|--------------|
| count | 10000.000000     | 10000.000000 | 10000.000000 |
| mean  | 37.778167        | 92.548900    | 84.976600    |
| std   | 1.769470         | 4.611197     | 26.305841    |
| min   | 35.000000        | 85.000000    | 40.000000    |
| 25%   | 36.111111        | 88.000000    | 63.000000    |
| 50%   | 37.777778        | 93.000000    | 85.000000    |
| 75%   | 39.444444        | 97.000000    | 108.000000   |
| max   | 40.555556        | 100.000000   | 130.000000   |

Figure 11: Summary Statistics of Selected Vital Signs

Additionally, the dataset's structure was examined using the 'df.info()' method to assess data types and identify any missing values. This step confirmed that the relevant features were correctly typed and free of missing data, further supporting the dataset's suitability for machine learning applications. (Figure 12)

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10002 entries, 0 to 10001
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Body Temperature 10000 non-null  float64
1   SpO2             10000 non-null  float64
2   Heart Rate       10000 non-null  float64
dtypes: float64(3)
memory usage: 234.6 KB
```

Figure 12: Data types and missing values assessment of the dataset

### 4.1.3 Labeling Thyroid Conditions

Thyroid anomalies were labeled based on established medical criteria derived from relevant literature and clinical guidelines. Three classes were defined as follows:

- Normal (Label 0): Represents healthy individuals without thyroid dysfunction.
- Hypothyroid (Label 1): Characterized by body temperature below 36.4°C, oxygen saturation (SpO<sub>2</sub>) less than 93%, and heart rate (HR) below 70 beats per minute (BPM).
- Hyperthyroid (Label 2): Defined by body temperature exceeding 37.3°C and heart rate above 100 BPM.

The resulting class distribution revealed a significant imbalance within the dataset. Normal cases constituted approximately 76% of the data (Figure 13), while hypothyroid and hyperthyroid cases represented only 18% and 4%, respectively. This imbalance poses challenges for model training and necessitates appropriate handling strategies to ensure robust classification performance.

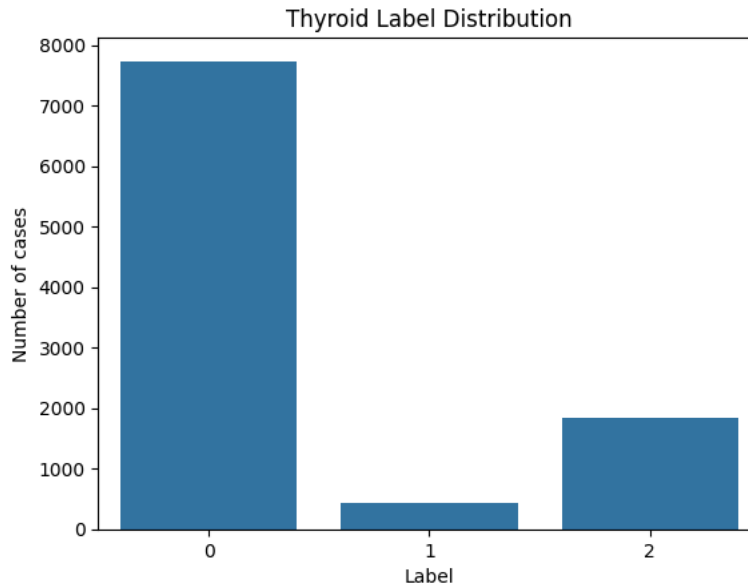


Figure 13: Bar chart of the Distribution of Thyroid Condition Labels in the Dataset

## 4.2 Preprocessing and Feature Selection

### 4.2.1 Data Preprocessing

To ensure data integrity and consistency, all rows containing missing values were removed using the ‘dropna()’ function from the Pandas library. This step was essential to eliminate incomplete records and to ensure that all selected samples included valid readings for body temperature, SpO<sub>2</sub>, and heart rate—key indicators used in labeling thyroid conditions. The removal of missing entries helped maintain data quality and minimized potential biases during the model training phase.

The dataset initially exhibited a substantial class imbalance, with the Normal class accounting for 7,734 samples, compared to only 1,832 for Hyperthyroid and 434 for Hypothyroid cases (Figure 13).

To mitigate this imbalance, two complementary strategies were employed. First, the Normal class was downsampled to 2,000 samples to reduce the disparity and limit overrepresentation (Figure 14.1). Then, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training dataset (Figure 14.2). SMOTE addresses imbalance by generating synthetic samples for the minority classes—Hypothyroid and Hyperthyroid—through interpolation between existing instances and their k-nearest neighbors. This approach ensures a more balanced distribution during training, improving the model’s sensitivity to minority classes and reducing bias toward the dominant class.

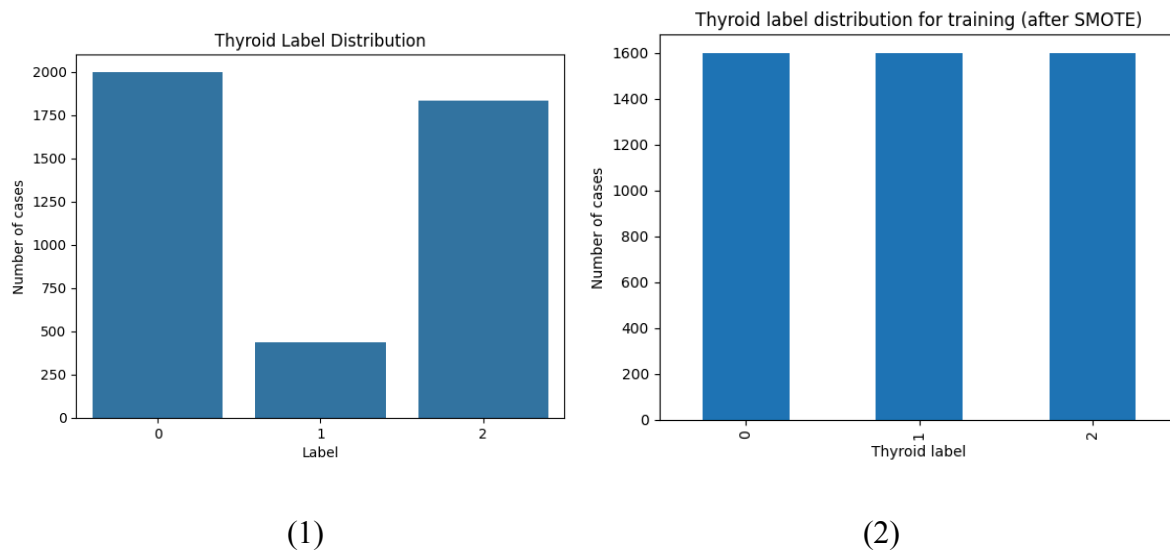


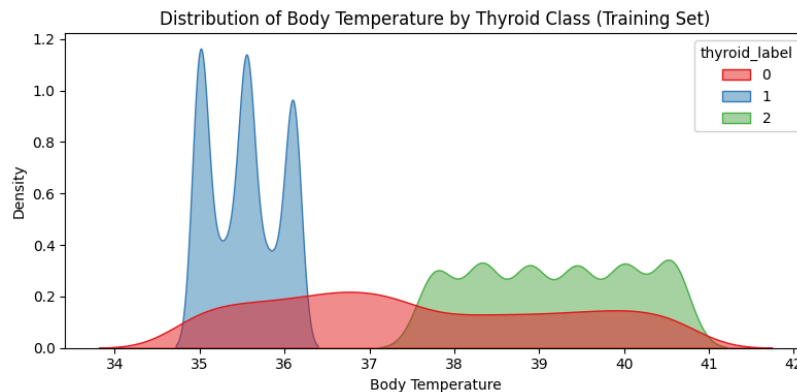
Figure 14: (1) Bar Chart of the Label Distribution after Downsampling. (2) Bar Chart of the Label Distribution after SMOTE.

#### 4.2.2 Data Visualization

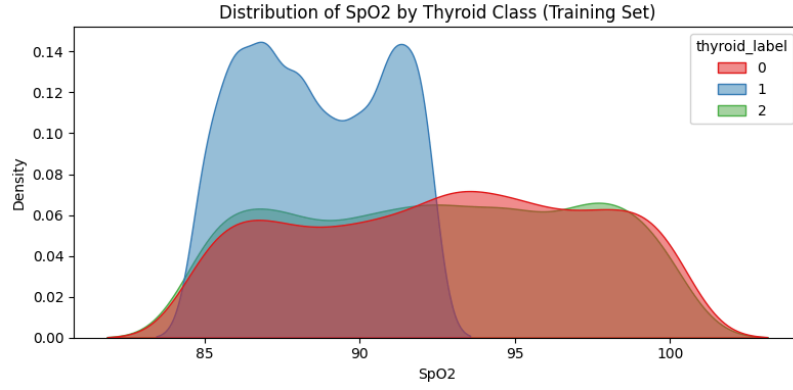
To gain deeper insights into the characteristics of the dataset and the relationship between features and thyroid conditions, visual exploration was performed. The distribution of body temperature across the three classes (0: Normal, 1: Hypothyroid, 2: Hyperthyroid) revealed distinct patterns aligned with clinical expectations (Figure 15). Specifically, individuals labeled as Hypothyroid tended to show lower body temperatures, consistent with symptoms such as cold intolerance. In contrast, those categorized as Hyperthyroid displayed elevated body temperatures, reflecting typical symptoms such as heat intolerance. These observations support the use of body temperature as a physiologically grounded, non-invasive indicator of thyroid dysfunction.

In addition to temperature, the distribution of SpO<sub>2</sub> (oxygen saturation) levels revealed meaningful distinctions among classes. The Hypothyroid group exhibited consistently lower SpO<sub>2</sub> values relative to other groups, aligning with documented impairments in respiratory function often associated with reduced thyroid hormone levels. This deviation further reinforces SpO<sub>2</sub> as a supportive indicator for hypothyroid detection.

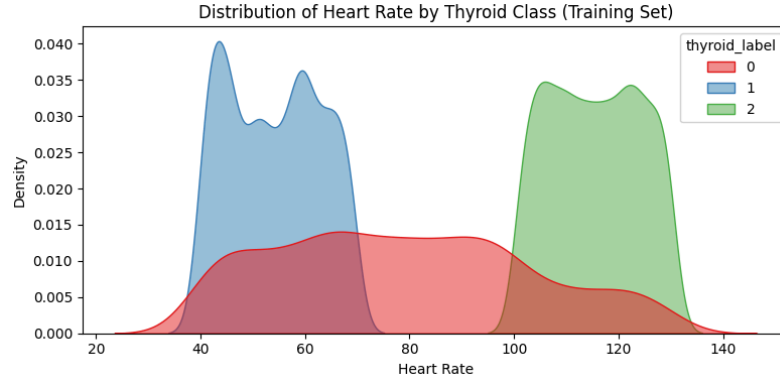
Similarly, heart rate (BPM) proved to be a key discriminative feature. Individuals in the Hypothyroid class generally showed bradycardic tendencies (lower heart rates), while those in the Hyperthyroid class exhibited tachycardia (elevated heart rates). The Normal class (euthyroid individuals) demonstrated a broader and more evenly distributed heart rate range, reflecting expected physiological variability. These feature distributions collectively highlight the diagnostic relevance of body temperature, SpO<sub>2</sub>, and heart rate in the context of thyroid dysfunction.



(a)



(b)



(c)

Figure 15: Feature Distributions Across Thyroid Classes:

(a) Body Temperature, (b) SpO<sub>2</sub>, (c) Heart Rate.

Although feature selection was guided by clinical relevance, the correlation matrix provides additional insight into the statistical associations between physiological variables and thyroid class labels (Figure 16). Body temperature and heart rate exhibited moderate positive correlations with the thyroid condition label, reinforcing their diagnostic importance for both hypothyroid and hyperthyroid states. These findings align with known clinical manifestations, such as elevated heart rate and temperature in hyperthyroidism, and their reduction in hypothyroidism.

Notably, SpO<sub>2</sub> demonstrated a weak overall correlation with the thyroid label. However, this low global correlation belies its specific diagnostic value: SpO<sub>2</sub> remains particularly relevant for identifying hypothyroid patients, who tend to present with reduced oxygen saturation. Thus, while SpO<sub>2</sub> may not contribute broadly across all classes, it plays a targeted and clinically meaningful role in the detection of hypothyroidism.

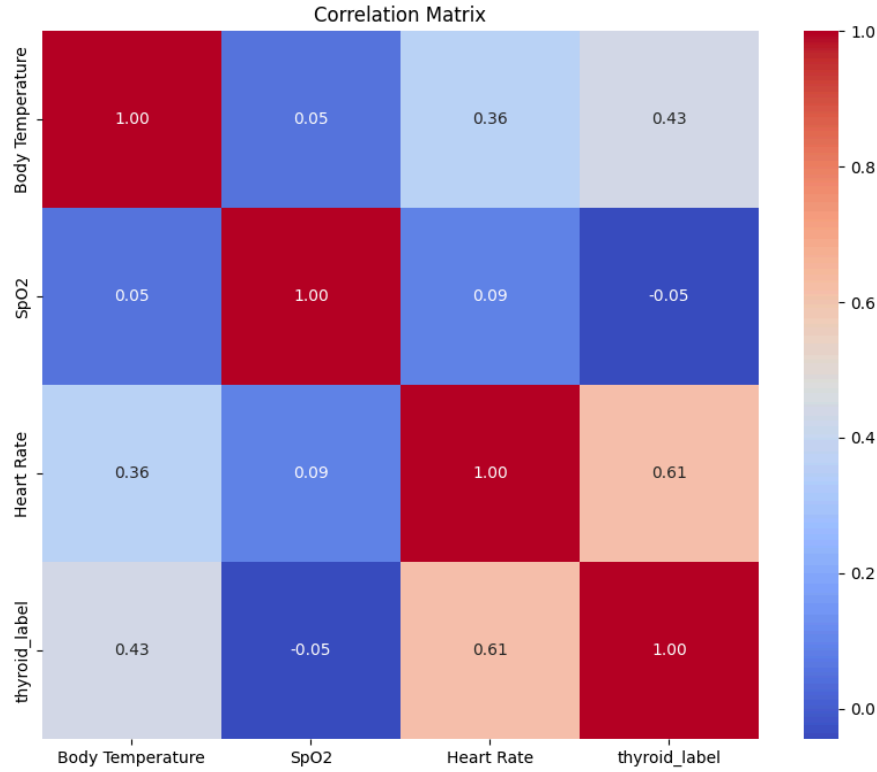


Figure 16: Correlation Heatmap of Selected Physiological Features and Thyroid Class Label.

#### 4.2.3 Data Split for Model Training

To assess model performance objectively, the dataset was partitioned into training and testing subsets using an 80/20 stratified split. Stratification was applied to ensure that each subset maintained the original distribution of thyroid condition classes. This approach was crucial for minimizing bias, especially given the initial class imbalance.

Feature scaling was employed selectively, depending on the requirements of the classification algorithms. Standardization techniques were used to ensure that features shared a common scale, typically with zero mean and unit variance. This step is particularly important for models that rely on distance metrics or gradient-based optimization, such as multilayer perceptrons (MLP) and support vector machines (SVM) with radial basis function (RBF) kernels.

In contrast, tree-based models like Random Forest and XGBoost are inherently insensitive to the scale of features. Therefore, scaling was not applied when training such models. By tailoring preprocessing to each model's characteristics, the study aimed to preserve performance integrity while enabling fair comparison across classifiers.

## 4.3 Model Architecture and Training Workflow

### *4.3.1 Model Architecture:*

To accurately detect thyroid-related conditions using physiological signals—namely body temperature, oxygen saturation (SpO<sub>2</sub>), and heart rate—a diverse set of machine learning classifiers was selected. This variety ensures a comprehensive evaluation of both linear and non-linear relationships within the data, enhancing predictive performance and interpretability.

Logistic Regression was employed as a baseline due to its simplicity and interpretability. It is particularly useful for assessing the linear influence of each physiological parameter on thyroid function and offers a low risk of overfitting, making it a suitable starting point for comparison.

To address the presence of non-linear patterns typical in biomedical data, a Support Vector Machine (SVM) with a radial basis function (RBF) kernel was included. The RBF kernel enables the model to map input features into a higher-dimensional space, allowing it to uncover complex, non-linear decision boundaries that may better capture physiological variations.

A Random Forest classifier was selected for its strength as an ensemble method capable of modeling non-linear interactions while also providing measures of feature importance. This capacity is particularly valuable in clinical contexts, as it helps to identify the most influential biomarkers in the classification process.

The XGBoost classifier was chosen for its robust performance on structured medical datasets. It effectively handles missing values and class imbalances, and its built-in regularization mechanisms help mitigate overfitting. Additionally, XGBoost is capable of modeling intricate interactions between features, making it well-suited to the complexities of physiological data.

Finally, a Multi-Layer Perceptron (MLP) classifier was included to exploit the strengths of deep learning. Although less interpretable, the MLP can model highly non-linear and abstract feature interactions, potentially revealing latent patterns that simpler models may overlook.

Overall, the integration of both linear and non-linear models provides a strategic framework for comparing transparency with predictive power. This approach supports a more nuanced understanding of the dataset and informs the selection of the most effective model for reliable biomedical prediction.

### 4.3.2 Training Configuration & workflow

Each model was trained using tailored hyperparameters to ensure optimal performance. The training process involved tuning key parameters via grid search or relying on well-established defaults when appropriate. The table below summarizes the configuration used:

#### Logistic Regression (LR)

- **Library:** sklearn.linear\_model.LogisticRegression
- **Parameters:** max\_iter=1000
- **Preprocessing:** SMOTE applied to training data for class balancing.
- **Input:** X\_train\_resampled, y\_train\_resampled
- **Purpose:** Baseline linear classifier, interpretable

#### Support Vector Machine (SVM, RBF Kernel)

- **Library:** sklearn.svm.SVC
- **Parameters:** kernel='rbf', C=100, gamma=0.01
- **Preprocessing:** SMOTE applied.
- **Input:** X\_train\_resampled, y\_train\_resampled
- **Purpose:** Non-linear decision boundaries via RBF kernel.

#### Random Forest (RF)

- **Library:** sklearn.ensemble.RandomForestClassifier
- **Parameters:** n\_estimators=100, random\_state=42
- **Preprocessing:** SMOTE applied.
- **Input:** X\_train\_resampled, y\_train\_resampled
- **Purpose:** Ensemble learning with feature importance.

#### XGBoost

- **Library:** xgboost.XGBClassifier
- **Parameters:** objective='multi:softmax', num\_class=3
- **Preprocessing:** SMOTE applied.
- **Input:** X\_train\_resampled, y\_train\_resampled
- **Purpose:** High-performance gradient boosting for multi-class classification.

#### Multi-Layer Perceptron (MLP)

- **Library:** sklearn.neural\_network.MLPClassifier
- **Parameters:**
  - hidden\_layer\_sizes=(32,)



- activation='relu'
- solver='adam'
- max\_iter=500
- random\_state=42
- **Preprocessing:**
  - Data normalized with StandardScaler
  - SMOTE applied before scaling
- **Input:** X\_train\_scaled, y\_train\_resampled
- **Purpose:** Deep learning for non-linear feature interactions

## 4.4 Evaluation Metrics and Results

### *4.4.1 Evaluation Metrics Used*

To evaluate model performance, a set of standard classification metrics was employed, offering both general and class-specific insights:

-**Accuracy** measures the overall proportion of correctly classified instances across all classes.

- **Precision** quantifies the proportion of true positives among all instances predicted as positive for a given class, reflecting the model's reliability in positive predictions.

-**Recall** (or sensitivity) represents the proportion of true positives among all actual positive cases, indicating the model's ability to capture relevant instances.

-**F1-score**, defined as the harmonic mean of precision and recall, balances these two aspects, particularly when there is a trade-off between them.

-**Confusion matrix** presents a detailed tabular comparison between predicted and actual labels, enabling a deeper understanding of misclassification patterns across classes.

These metrics collectively offer a comprehensive assessment of model performance, which is especially critical in multi-class, imbalanced medical classification tasks. They ensure that both prediction accuracy and the model's sensitivity to each thyroid condition are thoroughly evaluated.

### *4.4.2 Models Performance*

To begin the evaluation process, all models were assessed using a test set derived from the initial train-test split, where 80% of the data was allocated for training and 20% for testing. This internal evaluation provides a consistent baseline to compare model performance under

controlled conditions. Notably, several models achieved perfect or near-perfect accuracy (i.e., 1.0) on this internal test set, suggesting a strong initial fit to the data (Figure 16). However, such results may also indicate potential overfitting, especially in imbalanced datasets, highlighting the need for further validation.

```
LogisticRegression :
```

|              |                                   |
|--------------|-----------------------------------|
|              | [[332 30 38]                      |
|              | [ 1 86 0]                         |
|              | [ 24 0 343]]                      |
|              | precision recall f1-score support |
| 0            | 0.93 0.83 0.88 400                |
| 1            | 0.74 0.99 0.85 87                 |
| 2            | 0.90 0.93 0.92 367                |
| accuracy     | 0.89 854                          |
| macro avg    | 0.86 0.92 0.88 854                |
| weighted avg | 0.90 0.89 0.89 854                |

(a)

```
SVM (kernel RBF) :
```

|              |                                   |
|--------------|-----------------------------------|
|              | [[370 12 18]                      |
|              | [ 0 87 0]                         |
|              | [ 4 0 363]]                       |
|              | precision recall f1-score support |
| 0            | 0.99 0.93 0.96 400                |
| 1            | 0.88 1.00 0.94 87                 |
| 2            | 0.95 0.99 0.97 367                |
| accuracy     | 0.96 854                          |
| macro avg    | 0.94 0.97 0.95 854                |
| weighted avg | 0.96 0.96 0.96 854                |

(b)

```
Random Forest :
```

|              |                                   |
|--------------|-----------------------------------|
|              | [[400 0 0]                        |
|              | [ 0 87 0]                         |
|              | [ 0 0 367]]                       |
|              | precision recall f1-score support |
| 0            | 1.00 1.00 1.00 400                |
| 1            | 1.00 1.00 1.00 87                 |
| 2            | 1.00 1.00 1.00 367                |
| accuracy     | 1.00 854                          |
| macro avg    | 1.00 1.00 1.00 854                |
| weighted avg | 1.00 1.00 1.00 854                |

(c)

```
XGBClassifier :
```

|       |   |    |       |
|-------|---|----|-------|
| [[400 | 0 | 0] |       |
| [     | 0 | 87 | 0]    |
| [     | 0 | 0  | 367]] |

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 1.00      | 1.00   | 1.00     | 400     |
| 1            | 1.00      | 1.00   | 1.00     | 87      |
| 2            | 1.00      | 1.00   | 1.00     | 367     |
| accuracy     |           |        | 1.00     | 854     |
| macro avg    | 1.00      | 1.00   | 1.00     | 854     |
| weighted avg | 1.00      | 1.00   | 1.00     | 854     |

(d)

```
MLP Classifier :
```

|       |   |    |       |
|-------|---|----|-------|
| [[399 | 1 | 0] |       |
| [     | 0 | 87 | 0]    |
| [     | 0 | 0  | 367]] |

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 1.00      | 1.00   | 1.00     | 400     |
| 1            | 0.99      | 1.00   | 0.99     | 87      |
| 2            | 1.00      | 1.00   | 1.00     | 367     |
| accuracy     |           |        | 1.00     | 854     |
| macro avg    | 1.00      | 1.00   | 1.00     | 854     |
| weighted avg | 1.00      | 1.00   | 1.00     | 854     |

(e)

Figure 16: Results of different models:

(a) Logistic Regression (b) SVM (c) Random Forest (d) XGB Classifier (e) MLP Classifier

The exceptionally high scores observed during internal testing raised concerns regarding potential overfitting or limited generalization capabilities, particularly in light of the class imbalance and the use of synthetic oversampling (SMOTE) during training. To determine whether the models were genuinely learning meaningful patterns rather than overfitting to the training distribution, an additional evaluation was conducted using an external test set. This external validation provided a more realistic measure of model performance by simulating real-world deployment scenarios, where new patient data may differ significantly from the original training data. Such an approach is essential for verifying the robustness and generalizability of machine learning models in clinical applications.

To assess the generalization capability of the trained models, an external test set was employed, sourced from a publicly available synthetic dataset on Kaggle [32]. This dataset contains longitudinal synthetic health metrics for 100 patients recorded at 10-minute intervals over the span of one year. It includes various physiological signals such as heart rate, blood pressure, SpO<sub>2</sub>, respiration rate, body temperature, blood glucose, activity level, and sleep patterns.

Importantly, the dataset incorporates a wide range of simulated anomalies—such as sudden spikes, gradual drifts, periodic fluctuations, and sensor noise—to better mimic real-world clinical scenarios and variability in patient monitoring.

For external validation, we extracted three physiological features (Figure 17)—heart rate, SpO<sub>2</sub>, and body temperature—that matched those used in the original training dataset. By evaluating model performance on this external dataset, we aimed to validate the robustness and generalization of the models when applied to previously unseen, yet clinically relevant, variations in physiological signals.

| Index | Body Temperature   | SpO2              | Heart Rate        |
|-------|--------------------|-------------------|-------------------|
| 0     | 37.17986255884148  | 97.45569455913211 | 69.48072413607446 |
| 1     | 36.95597152616547  | 98.09147457405389 | 81.62695973561549 |
| 2     | 36.92434640333025  | 98.16674963969545 | 78.20835535713347 |
| 3     | 37.009841360854224 | 99.39763390590602 | 71.98646863100143 |
| 4     | 36.992326712009145 |                   | 77.93398886532175 |
| 5     | 37.256368965735845 | 82.55635951451161 | 70.94373777780615 |
| 6     | 37.480216130794936 | 97.85464943278564 | 70.3277948267451  |
| 7     | 37.32701801778968  | 98.26347406603449 | 79.82905695304541 |
| 8     | 36.934960062212156 | 98.858214257959   | 76.38134366857807 |
| 9     | 37.481357463669575 | 85.31060155061961 | 68.78225645624701 |
| 10    | 36.94748109432075  | 85.18500812601476 | 76.13403893265786 |
| 11    | 37.04522735927383  | 85.01971352987847 | 75.87909901646614 |
| 12    | 36.82249049819816  | 97.23649281363107 | 79.16971503848544 |
| 13    | 36.837339915832196 | 82.8677029893205  | 75.54612890602911 |
| 14    | 37.34818537190543  | 98.28947975902719 | 73.79855941379871 |
| 15    | 36.42956982454622  | 83.74465947084498 | 76.31681837828867 |
| 16    | 36.85850153450974  | 86.75671111766833 | 73.69950946144374 |

Figure 17: An overview of the dataset

Figure 18, present the performance metrics of each classification model evaluated on both the internal and external test sets. These visualizations summarize key indicators such as accuracy, precision, recall, and F1-score, allowing for a comparative assessment of how each model performs in controlled (internal) versus real-world-like (external) conditions. Particular attention is given to changes in performance between the two datasets, which highlight the models' ability—or limitations—in generalizing beyond the training data.

|                       |           |        |          |         |
|-----------------------|-----------|--------|----------|---------|
| Logistic regression : |           |        |          |         |
| [[1597    0    3]     |           |        |          |         |
| [   49 1551    0]     |           |        |          |         |
| [    1    0 1599]]    |           |        |          |         |
|                       | precision | recall | f1-score | support |
| 0                     | 0.97      | 1.00   | 0.98     | 1600    |
| 1                     | 1.00      | 0.97   | 0.98     | 1600    |
| 2                     | 1.00      | 1.00   | 1.00     | 1600    |
| accuracy              |           |        | 0.99     | 4800    |
| macro avg             | 0.99      | 0.99   | 0.99     | 4800    |
| weighted avg          | 0.99      | 0.99   | 0.99     | 4800    |

(a)

|                    |           |        |          |         |  |
|--------------------|-----------|--------|----------|---------|--|
| SVM (kernel RBF) : |           |        |          |         |  |
| [[1600 0 0]        |           |        |          |         |  |
| [1327 273 0]       |           |        |          |         |  |
| [1293 0 307]]      |           |        |          |         |  |
|                    | precision | recall | f1-score | support |  |
| 0                  | 0.38      | 1.00   | 0.55     | 1600    |  |
| 1                  | 1.00      | 0.17   | 0.29     | 1600    |  |
| 2                  | 1.00      | 0.19   | 0.32     | 1600    |  |
| accuracy           |           |        | 0.45     | 4800    |  |
| macro avg          | 0.79      | 0.45   | 0.39     | 4800    |  |
| weighted avg       | 0.79      | 0.45   | 0.39     | 4800    |  |

(b)

|                 |           |        |          |         |  |
|-----------------|-----------|--------|----------|---------|--|
| Random Forest : |           |        |          |         |  |
| [[1600 0 0]     |           |        |          |         |  |
| [ 190 1410 0]   |           |        |          |         |  |
| [1153 0 447]]   |           |        |          |         |  |
|                 | precision | recall | f1-score | support |  |
| 0               | 0.54      | 1.00   | 0.70     | 1600    |  |
| 1               | 1.00      | 0.88   | 0.94     | 1600    |  |
| 2               | 1.00      | 0.28   | 0.44     | 1600    |  |
| accuracy        |           |        | 0.72     | 4800    |  |
| macro avg       | 0.85      | 0.72   | 0.69     | 4800    |  |
| weighted avg    | 0.85      | 0.72   | 0.69     | 4800    |  |

(c)

|                 |           |        |          |         |  |
|-----------------|-----------|--------|----------|---------|--|
| XGBMClassifier: |           |        |          |         |  |
| [[1600 0 0]     |           |        |          |         |  |
| [ 0 1600 0]     |           |        |          |         |  |
| [1587 0 13]]    |           |        |          |         |  |
|                 | precision | recall | f1-score | support |  |
| 0               | 0.50      | 1.00   | 0.67     | 1600    |  |
| 1               | 1.00      | 1.00   | 1.00     | 1600    |  |
| 2               | 1.00      | 0.01   | 0.02     | 1600    |  |
| accuracy        |           |        | 0.67     | 4800    |  |
| macro avg       | 0.83      | 0.67   | 0.56     | 4800    |  |
| weighted avg    | 0.83      | 0.67   | 0.56     | 4800    |  |

(d)

|                  |           |        |          |         |  |
|------------------|-----------|--------|----------|---------|--|
| MLP Classifier : |           |        |          |         |  |
| [[1600 0 0]      |           |        |          |         |  |
| [ 589 1011 0]    |           |        |          |         |  |
| [ 719 0 881]]    |           |        |          |         |  |
|                  | precision | recall | f1-score | support |  |
| 0                | 0.55      | 1.00   | 0.71     | 1600    |  |
| 1                | 1.00      | 0.63   | 0.77     | 1600    |  |
| 2                | 1.00      | 0.55   | 0.71     | 1600    |  |
| accuracy         |           |        | 0.73     | 4800    |  |
| macro avg        | 0.85      | 0.73   | 0.73     | 4800    |  |
| weighted avg     | 0.85      | 0.73   | 0.73     | 4800    |  |

(e)

Figure 18: Results of different models:

(a) Logistic Regression (b) SVM (c) Random Forest (d) XGB Classifier (e) MLP Classifier

### Result Comparison:

After testing all models on the external dataset, we compared their performance visually using a plot (Figure 19) based on key evaluation metrics (accuracy, precision, recall, F1-score).

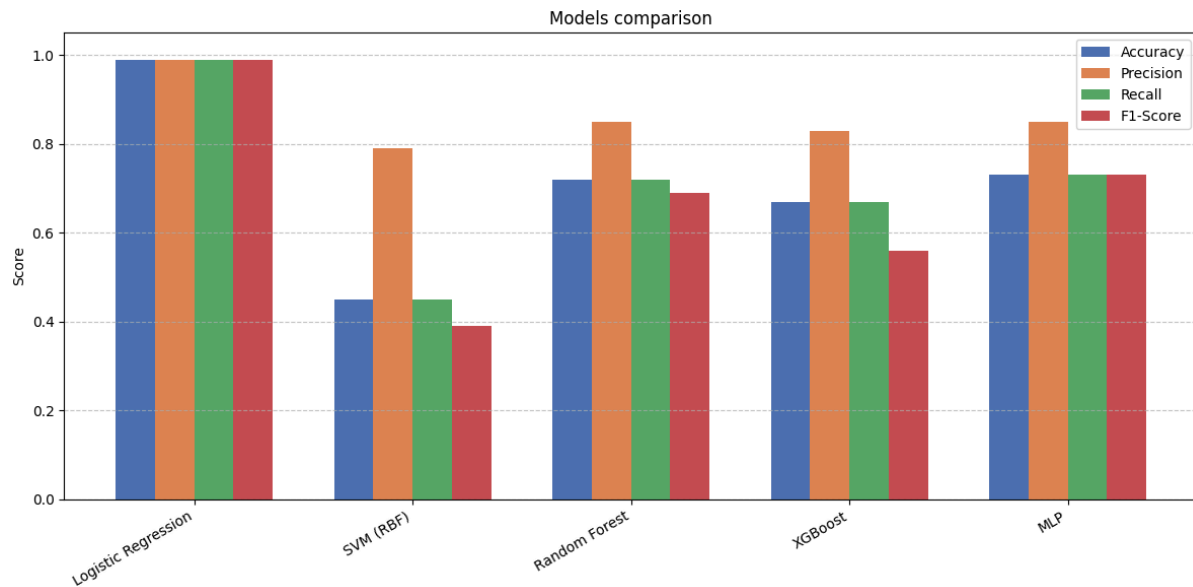


Figure 19: Bar Chart of Models Comaprison

#### *4.4.3 Model Selection and Justification*

After evaluating several models, Logistic Regression and MLP were selected as top performers for thyroid disorder prediction using vital signs.

Logistic Regression achieved high accuracy (99%) and excellent generalization, with the added benefit of interpretability, critical in medical contexts.

MLP provided robust performance (73% accuracy on external data), offering a flexible, non-linear architecture that captures complex feature interactions beyond linear boundaries.

We validated both models using real sensor data collected from our wearable prototype, which measures body temperature, SpO<sub>2</sub>, and heart rate. The models were deployed and tested directly on this live data, with predictions visualized in real time through a custom monitoring dashboard.

By combining these two models, and validating them with live data from our sensors, we confirmed their reliability in a real-world scenario.

## 5 Discussion and Future Perspectives:

### 5.1 Current Limitations:

Despite the promising outcomes of the proposed classification framework, several limitations constrain its clinical applicability and technical robustness. First, due to the absence of publicly available datasets containing vital signals explicitly associated with diagnosed thyroid disorders, proxy datasets were employed. This compromises the physiological specificity of the training data and limits the clinical interpretability of the model's outputs.

Furthermore, the models were developed using data that lack definitive thyroid-related annotations, restricting their generalizability to real-world thyroid patients. The current findings should therefore be interpreted as exploratory and indicative rather than conclusive.

In addition, the physiological signals used—particularly photoplethysmography (PPG) and skin temperature—are susceptible to motion artifacts and environmental interference. The sensing modality also does not integrate complementary biosignals such as electrocardiography (ECG), which could significantly enhance diagnostic accuracy.

On the hardware side, the wearable prototype remains constrained in terms of form factor, energy efficiency, and continuous monitoring capability. It is not yet optimized for long-term wear or daily use in practical settings.

Lastly, the system lacks full real-time integration. The absence of streaming protocols such as WebSocket or MQTT limits its deployment in remote healthcare infrastructures and real-time monitoring pipelines.

These limitations highlight essential directions for future refinement, which are addressed in the following section.

## 5.2 Future Extensions:

To address the current limitations and enhance the clinical and technical relevance of the system, several future directions are envisioned. One major avenue is the integration of electrocardiography (ECG) features into the data acquisition process. The addition of ECG signals may significantly improve both the sensitivity and specificity of thyroid-related anomaly detection, particularly in differentiating metabolic abnormalities from underlying cardiovascular issues.

Thyroid disorders exert distinct effects on cardiovascular electrophysiology that are reflected in electrocardiogram (ECG) parameters. In hypothyroidism, studies have reported a shorter corrected QT (QTc) interval, particularly among women, alongside a prolonged mean PR interval averaging  $165 \pm 25$  milliseconds, indicating delayed atrioventricular conduction. Conversely, hyperthyroidism is characterized by prolongation of the QT interval, as well as a reduction in both P-wave duration and QRS complex duration, with typical values around  $106.0 \pm 12.6$  milliseconds and  $88.5 \pm 9.6$  milliseconds, respectively. These alterations in ECG intervals are indicative of changes in atrial and ventricular conduction times and repolarization patterns associated with thyroid hormone imbalances. Therefore, accurate measurement of these intervals—PR, P-wave duration, QRS duration, and QT/QTc intervals—in reconstructed ECG signals derived from photoplethysmography (PPG) data is essential for detecting cardiovascular manifestations of thyroid dysfunction. [33]

In terms of system architecture, the adoption of real-time communication protocols such as WebSocket or MQTT is planned. These lightweight protocols will enable continuous data transmission to the GUI dashboard, thus supporting real-time monitoring and remote access. Such capabilities are especially valuable in telemedicine and long-term home-based patient follow-up.

From a hardware perspective, future iterations of the wearable device will prioritize miniaturization and operational optimization. This includes refining the printed circuit board (PCB) layout, implementing low-power operation modes, and introducing adaptive signal sampling strategies. These improvements are aimed at making the device suitable for extended daily use with minimal intrusiveness.



Another critical dimension involves advancing toward a context-aware health assessment framework. It is acknowledged that vital signs alone cannot replace a comprehensive clinical evaluation. The current system has been tested primarily under controlled, resting conditions, which limits its interpretability in scenarios involving emotional stress, physical exertion, or comorbid conditions.

To improve diagnostic reliability, a post-detection questionnaire system is also proposed. Designed in collaboration with medical students, this self-assessment tool will be presented through the dashboard interface following an anomaly alert. It will serve to contextualize physiological deviations by capturing subjective symptoms, thereby reducing the likelihood of false positives and enhancing clinical relevance.

Together, these planned enhancements aim to transition the system from a reactive, signal-driven anomaly classifier toward a more robust, context-aware, and patient-centered decision support tool with greater applicability in real-world healthcare settings.

## 6 Project Timeline:

Given the multidisciplinary nature of our project, which spans IoT, data science, embedded systems, and web development, it was essential to adopt a structured and organized workflow. Each axis required specific expertise and had interdependent deliverables.

To ensure efficient time management, task coordination, and progress tracking, we utilized a Gantt chart (Figure 20) as a project planning tool. This allowed us to clearly define milestones, allocate responsibilities, and synchronize hardware and software development phases.



Figure 20: Wearable device for Hypo-HyperThyroidism early diagnosis Gantt Chart

## 7 Conclusion:

This work presents a wearable-based health monitoring system tailored to the early classification of thyroid gland disorders using physiological signals such as PPG, SpO<sub>2</sub>, heart rate, and temperature. Through a machine learning-driven classification pipeline and an interactive dashboard interface, the system demonstrates the potential for continuous, non-invasive thyroid monitoring outside traditional clinical settings.

The comparative analysis of models and feature selection strategies confirms that vital signs carry meaningful patterns indicative of thyroid dysfunction, although their interpretation requires careful contextualization. While the system shows promising classification results under resting conditions, it remains an assistive tool rather than a diagnostic replacement.

Future development will focus on integrating richer biosignals (e.g., ECG), real-time communication protocols, and user-informed contextual layers such as post-detection questionnaires. These enhancements aim to improve the system's clinical relevance, user adoption, and decision-making accuracy in real-world applications.

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